

International Economic Development

ECON 4660/8666

Lecture 3: Theories of Economic Growth

I. Schumpeter's Model of Innovation

- Joseph Schumpeter (1934)
 - *The Theory of Economic Development*. Harvard U. press.
- Capitalist economy is a process of “creative destructions”. Each firm seeks to gain an advantage through innovation, e.g., a more attractive design, a new product, etc.
- As a result, the innovator destroys the monopoly power that its competitors gained because of earlier innovations.
- This continuous creation and destruction prevents permanent monopolies, and society experiences constant technological progress, which is the source of economic growth.

II. Rostow's stages of Growth

- Rostow Adviser to President Eisenhower on economic and foreign policy
 - “*Stages of Economic Growth: A Non-Communist Manifesto.*” 1960
 - “The Take-Off into Self-Sustained Growth” (Econ J. 46-1, 1956),
- 1. The traditional society.
 - Agriculture, self-sufficiency, persisted for centuries
- 2. The transitional stage (the pre-conditions for take-off into self-sustaining growth).
 - Surplus of labor in rural
 - more savings & investment, import of capital goods, development requires substantial investment in capital
 - Last for a number of decades

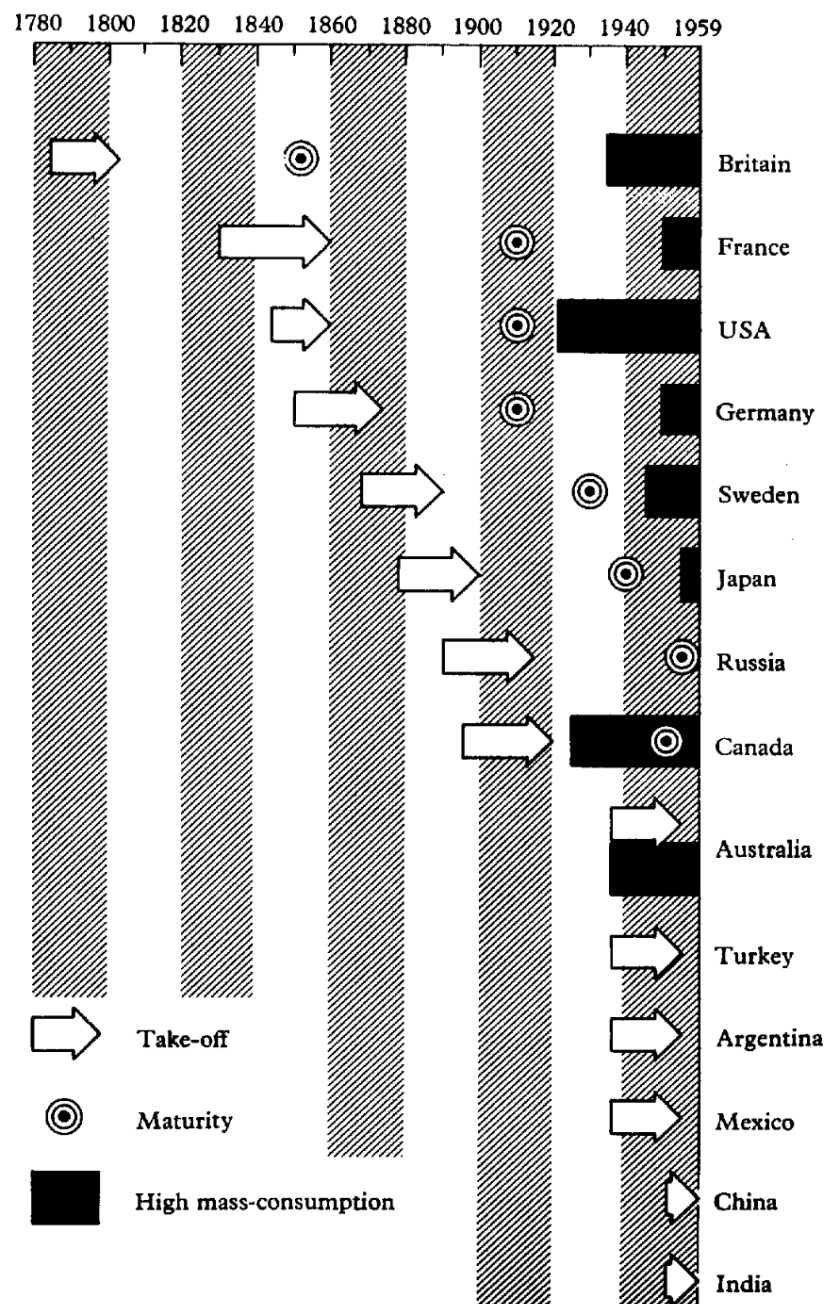


Chart of the stages of economic growth in selected countries. Note that Canada and Australia have entered the stage of high mass-consumption before reaching maturity. [By courtesy of the *Economist*.]

3. The take-off: growth rate accelerates, relatively brief

- Could be initiated by a revolution military coup in S. Korea in 1961 (Park Chung-Hee, 1961 military coup, 1963 elected president, till 1979);

But Khmer Rouge in Cambodia (“Red Cambodians”, followers of the Communist Party of Kampuchea, 1975 to 1979, Pol Pot, killed 3,300,000 people) and Khomeini in Iran, counter examples

- Technological breakthrough. The Industrial Revolution in England in the latter half of the 18th century
- Institutional development, such as commercial banks, thrift institutions, bond and stock markets, insurance companies, pension funds, etc. Channel domestic funds into investment
- Foreign investment
- Economic reforms, China

4. The drive to maturity: Sustained growth

The economy diversifies into new areas. Technological innovation is providing a diverse range of investment opportunities. The economy is producing a wide range of goods and services

5. The age of high mass consumption

The economy is moving towards mass consumption. Durable consumer goods industries flourish. Service sector becomes increasingly dominant. Economic security is provided.

6. The stage of pursuing quality of life

Improving quality of products. Increasing leisure time. Increasing education, healthcare, social welfare expenditures.

Buddenbrooks dynamics

In Thomas Mann's novel of Buddenbrooks dynamics (1901, German), there are three generations.

- The first sought money; the second, born to money, sought social and civic position; the third, born to comfort and family prestige, looked to the life of music.
- The phrase is designed to suggest, then, the changing aspirations of generations, as they place a low value on what they take for granted and seek new forms of satisfaction.
- The work led to a Nobel Prize in Literature for Mann in 1929.

Criticisms:

Many economists argue that Rostow's model does not set down the detailed nature of the pre-conditions for growth. In reality, policy makers are unable to clearly identify stages. Thus, not very helpful.

Rostow's view on Soviet Union:

Rostow saw in Russia “a nation surging under communism, into a long-delayed status as an industrial power of the first order.”

Policy impacts of Rostow's work:

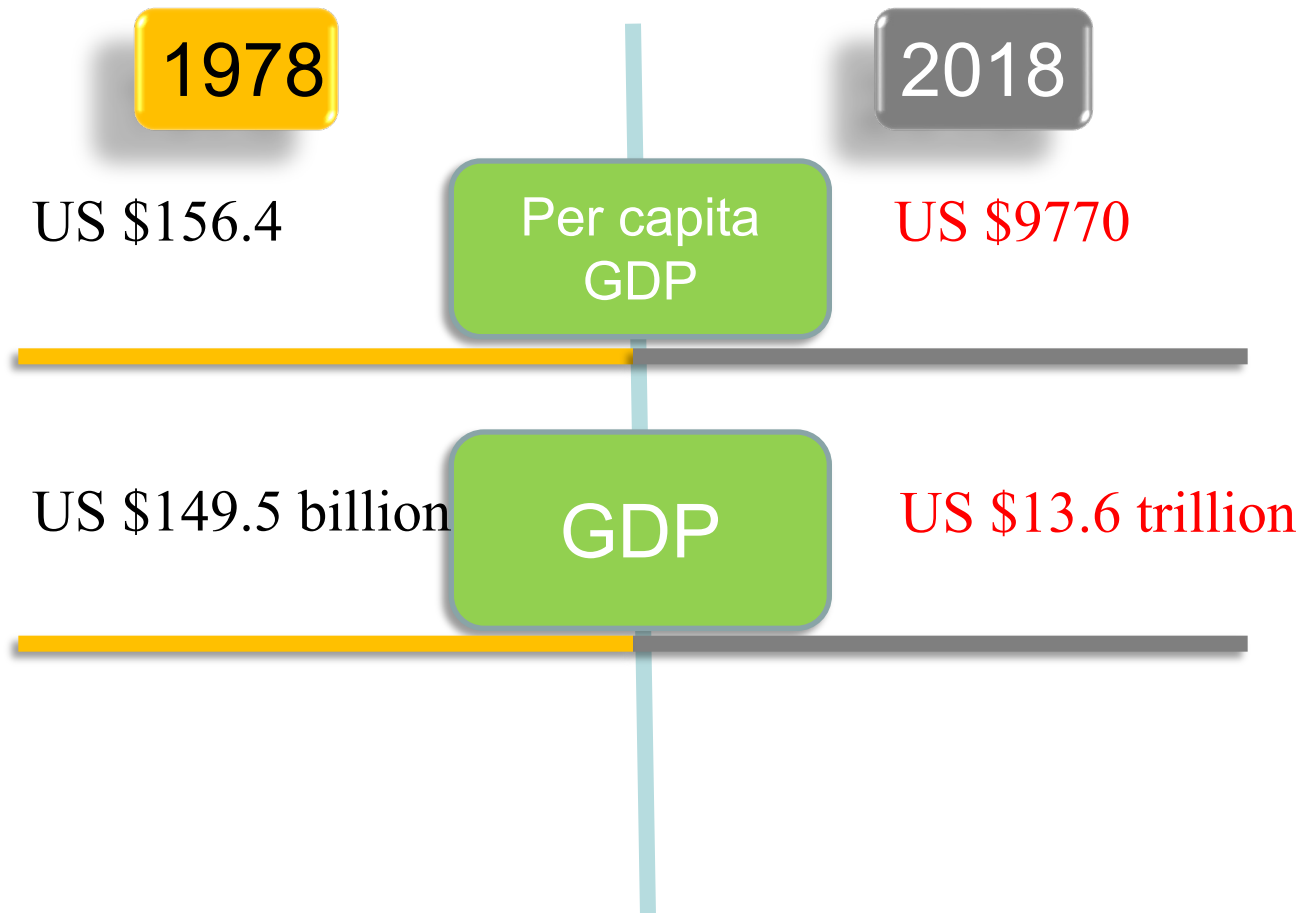
To prevent communist revolution, U.S. increased foreign aid in the late 1950s.

J.F.K. mentioned, new nations need help to reach self-sustaining growth for special reason, w/o exception, they are under communist pressure.

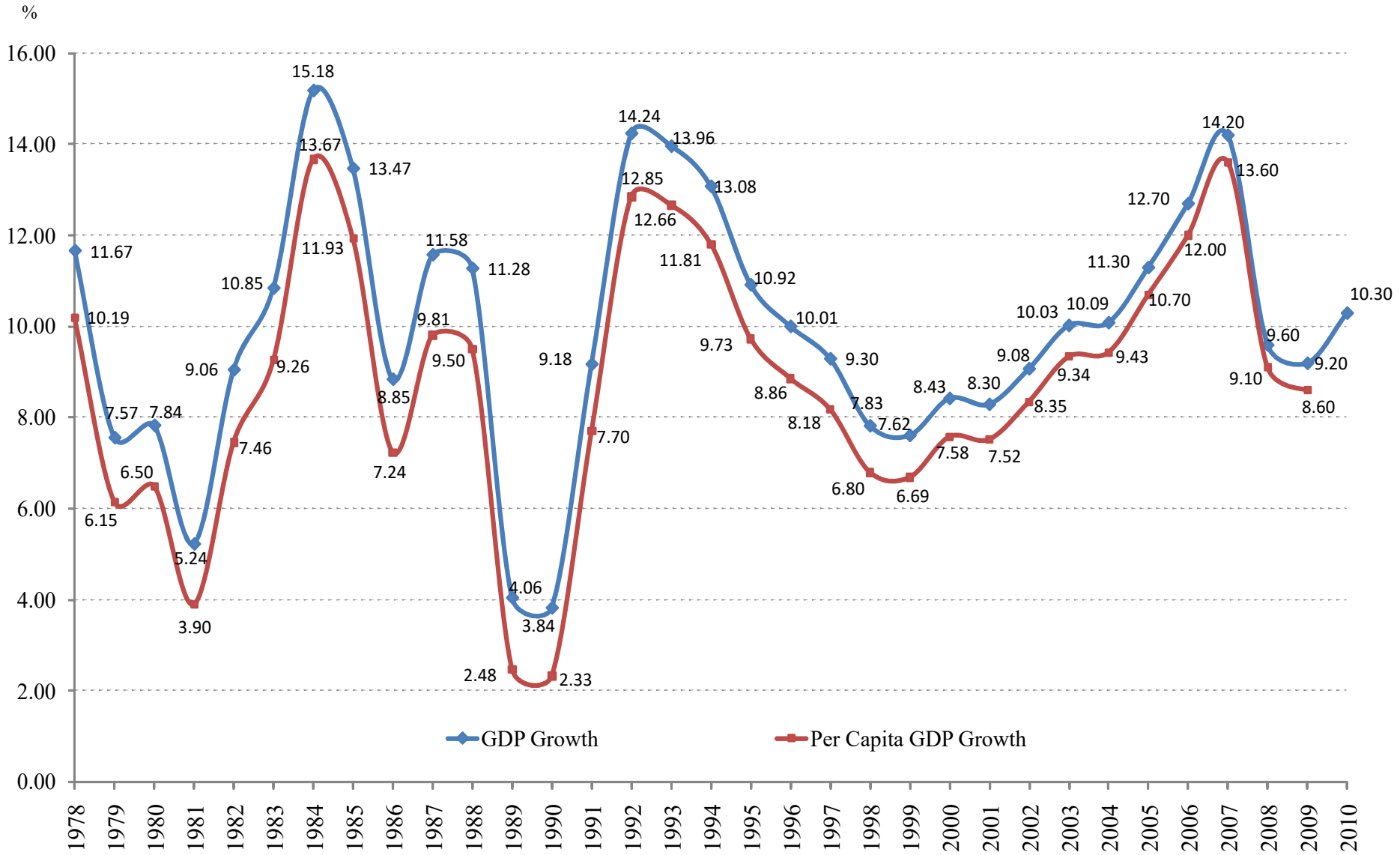
III. Economic Development in Selected Countries

1. China's economic growth

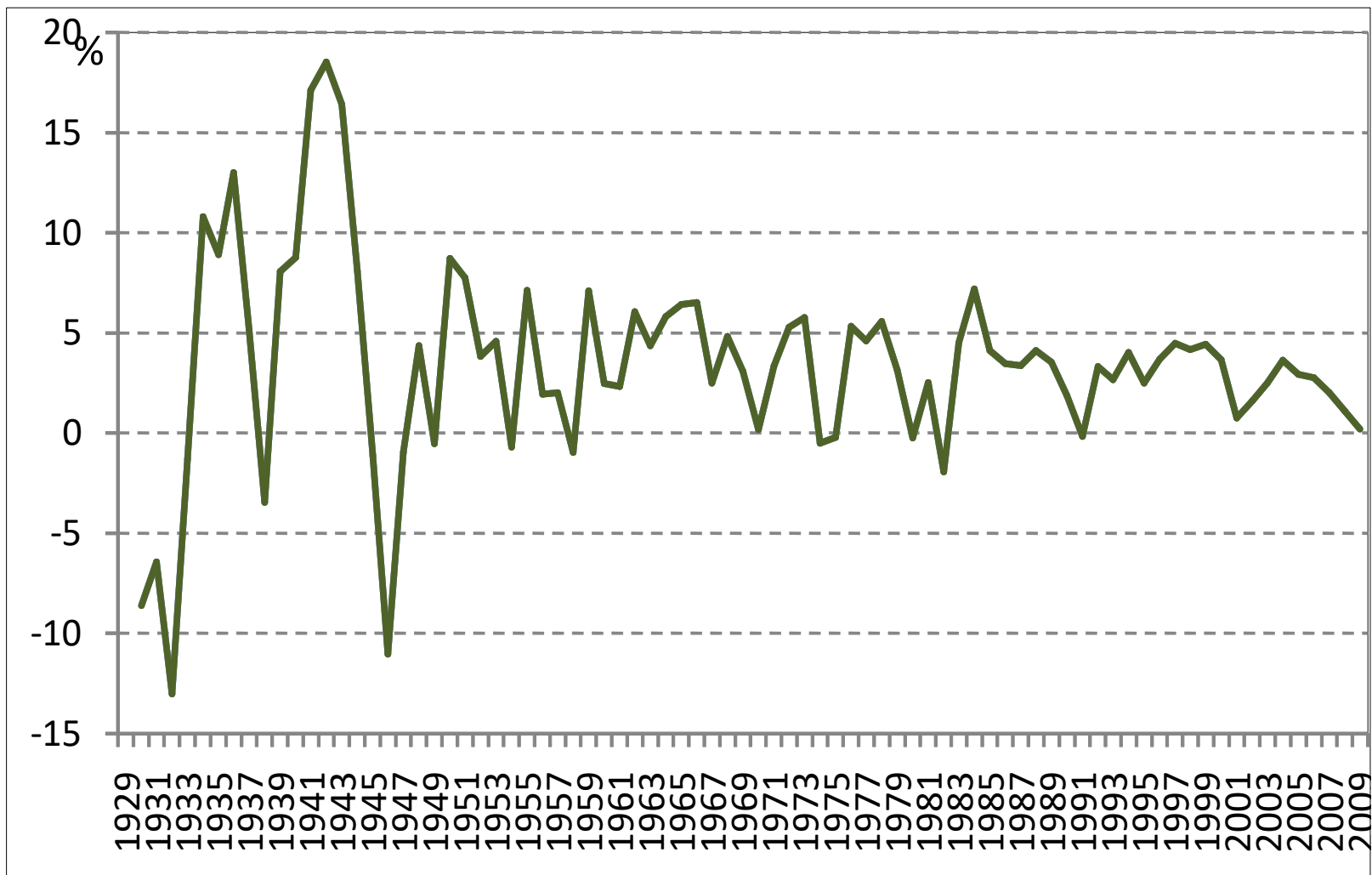
The economic reforms started in 1978 have transformed China from one of the poorest countries into an up-middle-income country.



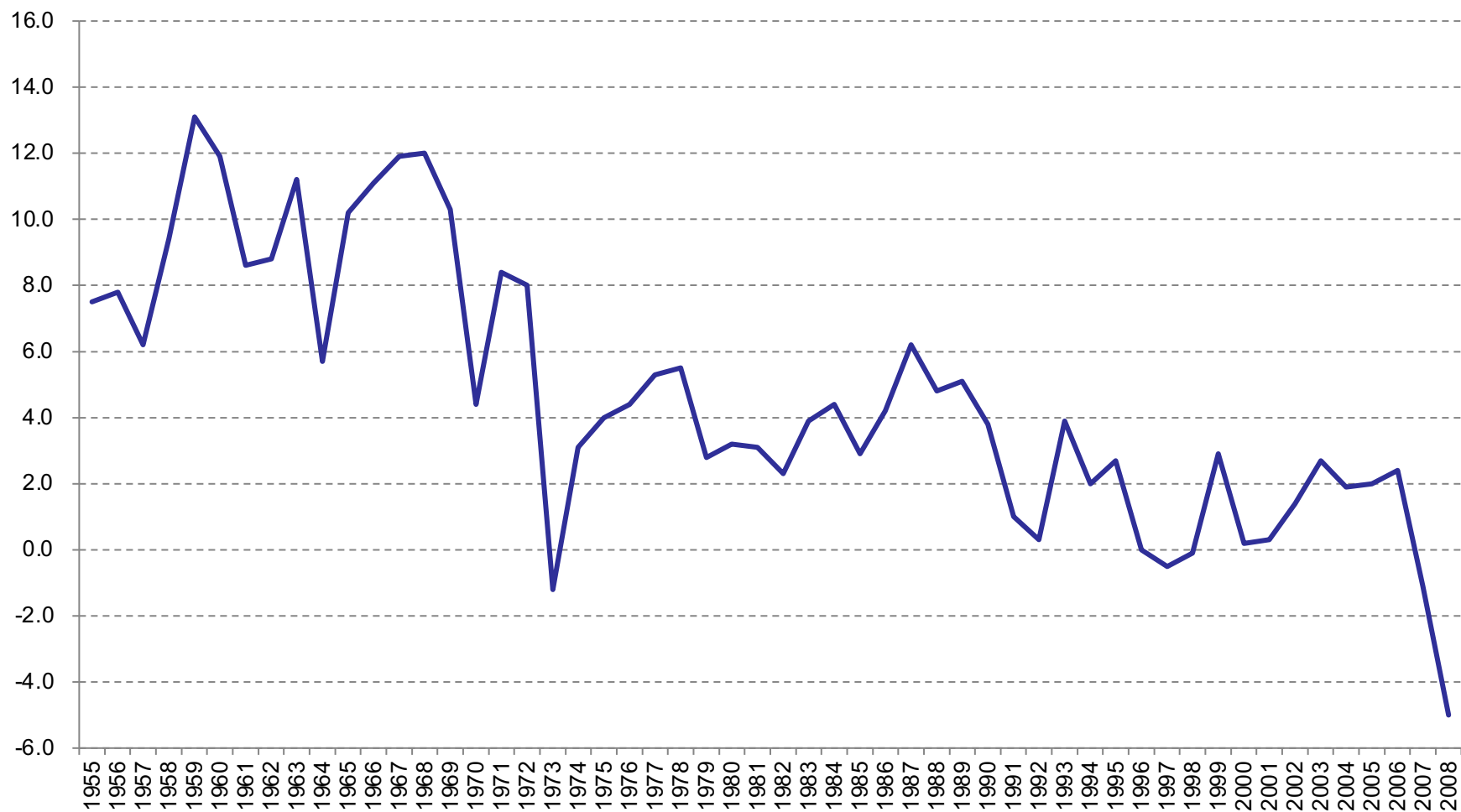
China's GDP Growth 1978-2010



2. US GDP Growth 1930-2009



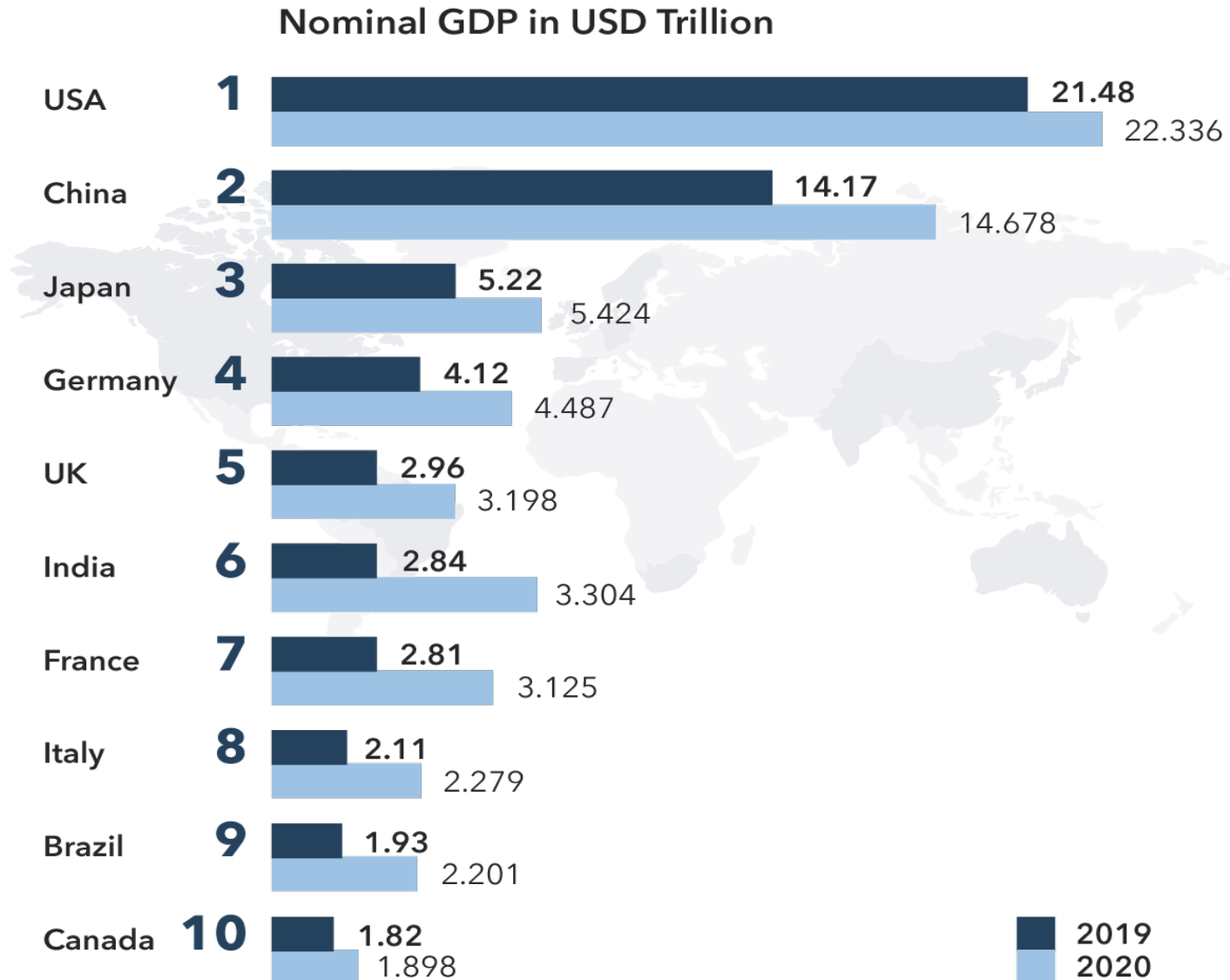
3. Japan's GDP Growth (1955-2009)



4.

TOP 10

The World's Biggest Economies for 2019 and 2020



5. Wealth of Nations 2000 (\$per capita) WDR 2010

Countries	Total wealth	Produced capital and urban land	Intangible capital	Natural capital	Pasture land	Crop land	Pro tected areas	Non-timber forest re sources	Timber resources	Subsoil assets
Australia	371,031	58,179	288,686	24,167	5,590	4,365	1,421	551	748	11,491
Bangladesh	6,000	817	4,221	961	52	810	9	2	4	83
Belgium	451,714	60,561	388,123	3,030	2,161	575	0	20	254	20
Brazil	86,922	9,643	70,528	6,752	1,311	1,998	402	724	609	1,708
Canada	324,979	54,226	235,982	34,771	1,631	2,829	5,756	1,264	4,724	18,566
China	9,387	2,956	4,208	2,223	146	1,404	27	29	106	511
Ethiopia	1,965	177	992	796	197	353	167	16	63	0
Franc	468,024	57,814	403,874	6,335	2,091	2,747	1,026	77	307	87
German	496,447	68,678	423,323	4,445	1,586	1,176	1,113	39	263	269
Greece	236,972	28,973	203,445	4,554	573	3,424	57	101	82	318
India	6,820	1,154	3,738	1,928	192	1,340	122	14	59	201
Indonesia	13,869	2,382	8,015	3,472	50	1,245	167	115	346	1,549
Japan	493,241	150,258	341,470	1,513	316	710	364	56	38	28
S Korea	141,282	31,399	107,864	2,020	275	1,241	441	30	0	33
Malaysia	46,687	13,065	24,520	9,103	24	1,369	161	188	438	6,922
Mexico	61,872	18,959	34,420	8,493	721	1,195	176	128	199	6,075
Philippines	19,351	2,673	15,129	1,549	45	1,308	59	17	90	30
Russia	38,709	15,593	5,900	17,217	1,342	1,262	1,317	1,228	292	11,777
Sweden	513,424	58,331	447,143	7,950	1,676	1,120	1,549	908	2,434	263
Thailand	35,854	7,624	24,294	3,936	96	2,370	855	55	92	469
UK	408,753	55,239	346,347	7,167	1,291	583	495	14	44	4,739
US	512,612	79,851	418,009	14,752	1,665	2,752	1,651	238	1,341	7,106
World	95,860	16,850	74,998	4,011	536	1,496	322	104	252	1,302
Low inc	7,532	1,174	4,434	1,925	189	1,143	111	48	109	325
Middle inc	27,616	5,347	18,773	3,426	407	1,583	129	120	169	1,089
High inc (OECD)	439,063	76,193	353,339	9,531	1,552	2,008	1,215	183	747	3,825

6. Population and GNI in 2008 WDR 2010

Country	Population (million)	GNI (US\$ billion)	Per capita GNI (US\$)	GNI based on PPP (US\$ billion)	GNI based on PPP(US\$)
Australia	21	862.5	40,350	727.5	34,040
Canada	33	1,390.00	41,730	1,206.50	36,220
Chile	17	157.5	9,400	222.4	13,270
China	1,326	3,899.30	2,940	7,984.00	6,020
Ethiopia	81	22.7	280	70.2	870
Finland	5	255.7	48,120	189.5	35,660
France	62	2,702.2	42,250	2,134.40	34,400
Germany	82	3,485.70	42,440	2,952.40	35,940
India	1,140	1,215.50	1,070	3,374.90	2,960
Indonesia	228	458.2	2,010	875.1	3,830
Japan	128	4,879.20	38,210	4,497.70	35,220
S Korea	49	1,046.30	21,530	1,366.90	28,120
Malaysia	27	188.1	6,970	370.8	13,740
Mexico	106	1,061.40	9,980	1,517.20	14,270
Norway	5	415.3	87,070	279	58,500
Philippines	90	170.4	1,890	352.4	3,900
Russia	142	1,364.50	9,620	2,216.30	15,630
Singapore	5	168.2	34,760	232	47,940
Spain	46	1,456.50	31,960	1,418.70	31,130
Sweden	9	469.7	50,940	352	38,180
Thailand	67	191.7	2,840	403.4	5,990
UK	61	2,787.20	45,390	2,218.20	36,130
US	304	14,466.10	47,580	14,282.70	46,970
World	6,692	57,637.5	8,613	69,309	10,357

IV. The Harrod-Domar Model

- To grow, savings must be greater than replacement.
- Output (Y) produced by capital (K):

$$Y = \frac{K}{C}$$

Where C is a constant, indicating capital-output ratio ($C = \frac{K}{Y}$)

- Changes in output: $\Delta Y = \frac{\Delta K}{C}$

Growth rate of output: $\frac{\Delta Y}{Y} = \left(\frac{\Delta K}{Y}\right) \frac{1}{C}$

Defining $g \equiv \frac{\Delta Y}{Y}$, $g = \frac{S}{Y} \times \frac{1}{C} = s / C$

$\Delta K \equiv I = S$, where $s \equiv \frac{S}{Y}$

- This is the famous Harrod-Domar theory of economic growth, indicating that the growth rate is determined by the savings rate and capital-output ratio (which is assumed to be constant).

High savings rate, high growth, low capital use, high growth.

- Example: $C = 3$, $s = 9\%$

$$g = \frac{9\%}{3} = 3\%$$

- Incremental capital-output ratio (ICOR):

$$\text{increase in capital/ increase in output} = \frac{\Delta K}{\Delta Y}$$

- Example: $\Delta K = 3,000$, $\Delta Y = 1,000$

$$\text{ICOR} = \frac{3000}{1000} = 3$$

- ICOR = COR if COR is constant.

Capital-output ratio, C

$$\begin{aligned}\frac{K}{Y} = C &\Rightarrow Y = \frac{K}{C} \Rightarrow \Delta Y = \frac{\Delta K}{C} \\ &\Rightarrow C = \frac{\Delta K}{\Delta Y}\end{aligned}$$

– Example:

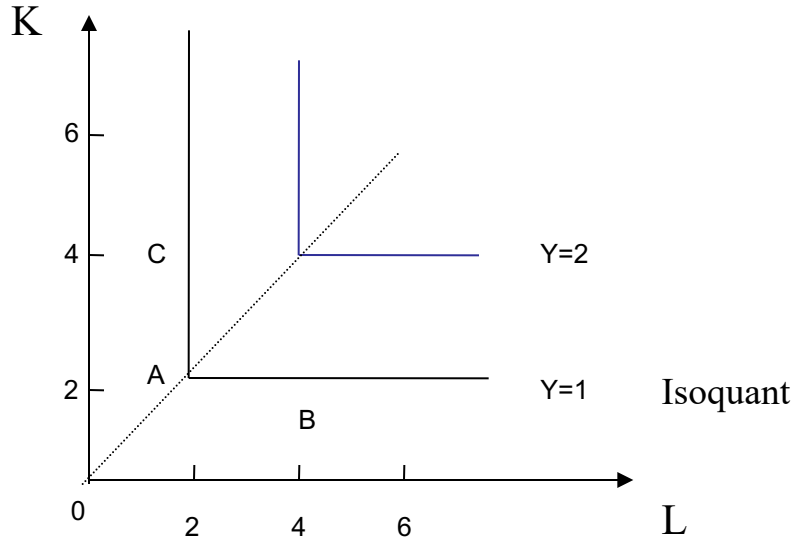
$$K = 3,000, Y = 1,000, \Delta K = 300, \Delta Y = 100$$

$$\frac{3000}{1000} = \frac{3000 + 300}{1000 + 100} = 3$$

- Production functions consistent with constant capital-output ratio:

- $Y = \frac{K}{C}$ (For Example, $C = 3$)

- $Y = \min\{aK, bL\}$, L is labor, $a=b=0.5$ constant
– Example:



	K	L	Y	
A	2	2	1	$\min\{0.5 \times 2, 0.5 \times 2\}$
B	2	4	1	$\min\{0.5 \times 2, 0.5 \times 4\}$
C	4	2	1	$\min\{0.5 \times 4, 0.5 \times 2\}$
A'	4	4	2	
B'	4	6	2	
C'	6	4	2	

- In the real world, capital-output ratio varies a lot.

	1980-1989	1990-1997
U.S.	3.6	2.8
Japan	4.6	6.5
Venezuela	6.7	2.8

V. Neoclassical Growth Theory

- Solow, MIT, Nobel prize, 1987
 - Adding labor input in the Harrod-Domar model, and assume that labor and capital are substitutable.

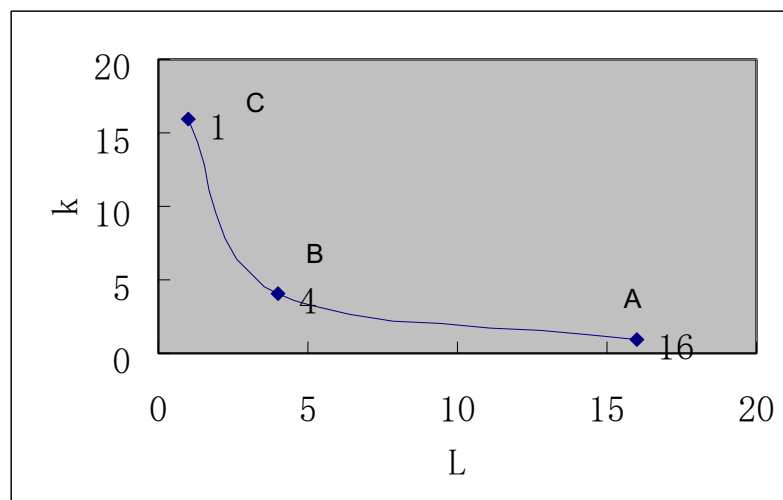
1. Production function $Y = f(K, L)$

- Example: $Y = \frac{1}{2} K^{0.5} L^{0.5}$

$$\frac{1}{2} 4^{0.5} 16^{0.5} = 2$$

Isoquants

	K	L	Y
A	1	16	2
B	4	4	2
C	16	1	2



- Output-Labor ratio

$$Y = F(K, L) \quad \frac{Y}{L} = F\left(\frac{K}{L}, 1\right)$$

Let $y \equiv \frac{Y}{L} \quad k \equiv \frac{K}{L}$

$$y = F(k, 1) = f(k)$$

2. Capital accumulation

$$\Delta K = sY - dK \quad \text{where } s = S/Y$$

s = saving rate, d = depreciation rate.

$$\Delta K/L = sy - dk \quad (1)$$

since $\Delta k = \Delta\left(\frac{K}{L}\right) = \frac{\Delta KL - \Delta LK}{L^2} = \frac{\Delta K}{L} - \frac{\Delta L}{L}k$

$$\Rightarrow \frac{\Delta K}{L} = \Delta k + \frac{\Delta L}{L}k \quad (2)$$

Let $n = \frac{\Delta L}{L}$

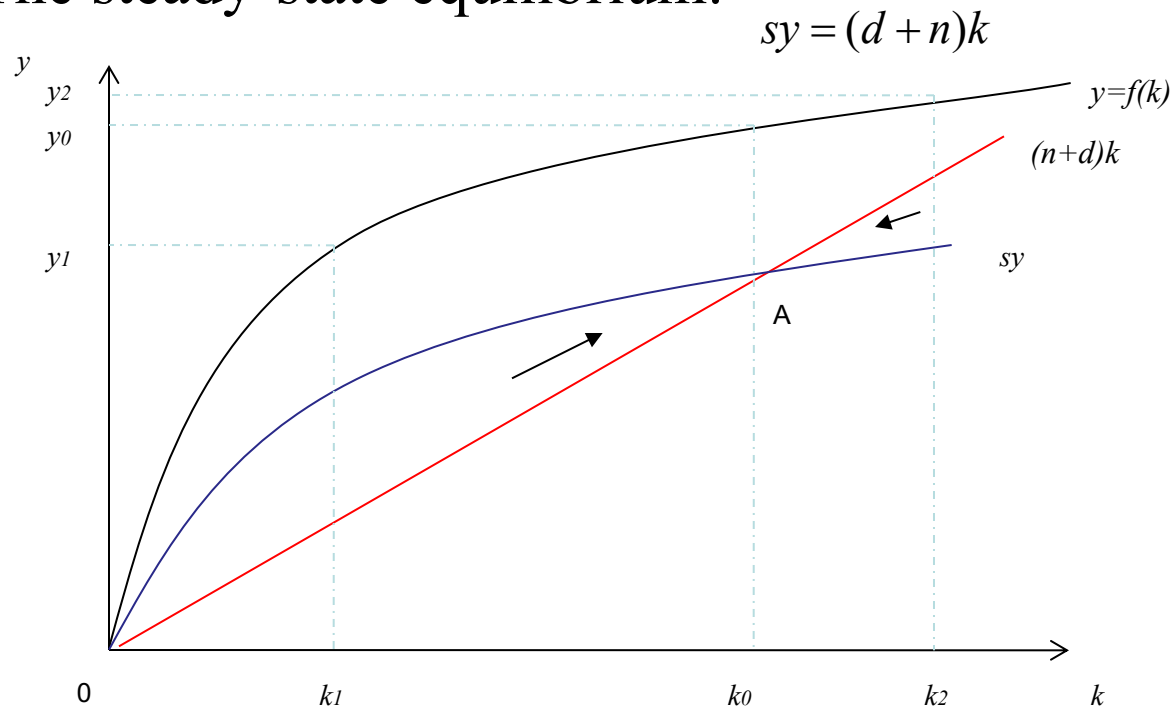
$$(1) \& (2) \Rightarrow \Delta k = sy - dk - nk$$

$$\Rightarrow \Delta k = sy - (d + n)k$$

- Change in capital-labor ratio
 - a. Positively related to per capita saving, sy
 - b. Negatively related to population growth, n
 - c. Negatively related to depreciation rate, d
- Capital deepening:
 - The process to increase k (i.e., $\Delta k > 0$) is called Capital deepening. Important for growth in transition.
- Capital widening:
 - The process to keep k constant (i.e., $\Delta k = 0$) is called Capital widening.

3. The steady-state equilibrium:

$$(\Delta k = 0, \Delta y = 0)$$



- k_0 : steady-state capital-labor ratio;
- y_0 : steady-state output-labor ratio.
- New saving = new capital needed for the growth of workforce & depreciation.
- The growth rate of output per worker, $\frac{Y}{L}$, and the growth rate of capital per worker $\frac{K}{L}$, are 0.
- Growth rate of $Y = \text{Growth rate of } L$.

$$\Delta\left(\frac{Y}{L}\right) = 0$$

$$\Delta\left(\frac{Y}{L}\right) = \frac{\Delta YL - \Delta LY}{L^2} = 0$$

$$\implies \Delta YL - \Delta LY = 0$$

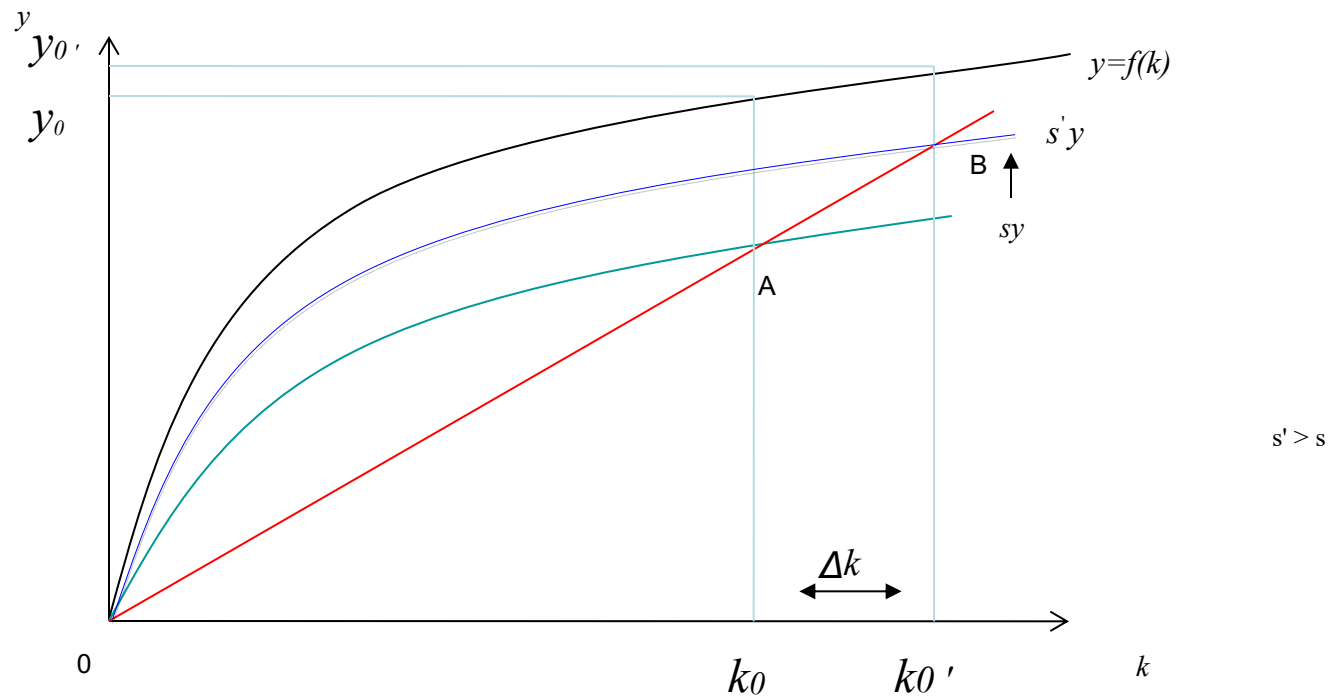
$$\implies \frac{\Delta Y}{Y} - \frac{\Delta L}{L} = 0$$

$$\implies \frac{\Delta Y}{Y} = \frac{\Delta L}{L}$$

- An implication of Solow model:
 - When countries have the same Steady State level of y , poor countries are expected to grow faster than rich countries and eventually catch up.
 - Think about the case where one country's income is at y_1 and the other country's income is at y_0 . The first country's per capita income will not grow, while the second country's income will grow.

See the previous picture.

4. An increase in the savings rate



$s \uparrow, k \uparrow, y \uparrow$ in transition, no growth in S.S.

- Savings rate does not affect S.S growth rate
- Facts: Investment can explain growth of East Asian industrialized economies, but not others.

Example:

- Nigeria & Hong Kong

$\frac{K}{L} \uparrow$ by over 250% from 1960-1985.

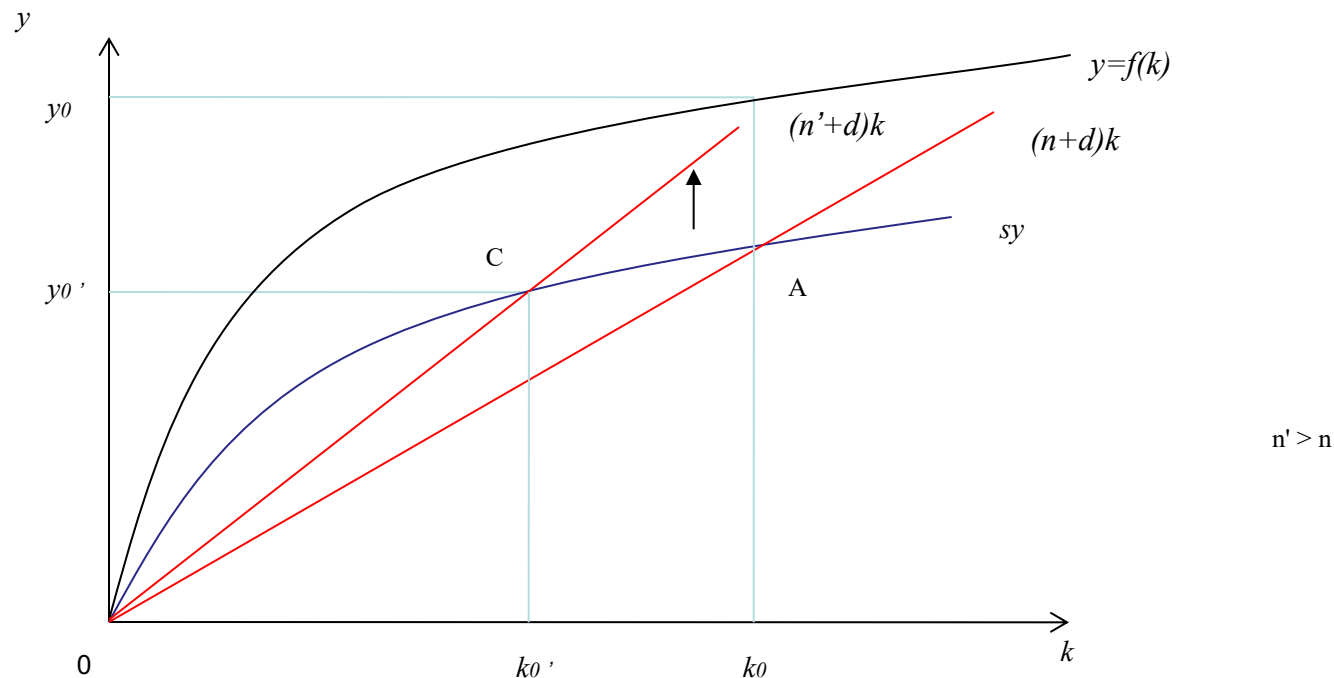
$\frac{Y}{L} \uparrow$ by 12% & 328% respectively.

- Gambia & Japan

$\frac{K}{L} \uparrow$ by over 500% from 1960-1985.

$\frac{Y}{L} \uparrow$ by 2% & 260% respectively.

5. An increase in the labor force growth



$n \uparrow, k \downarrow, y \downarrow$ in transition

- n does not affect the S.S. growth rate. S.S. growth is 0.

Note: a. Two countries with same income level may have different growth rates in transition if they have different S.S. levels of income (China, Indonesia).

S.S. income level is determined by saving rate, population growth rate, etc.

b. Two countries with the same S.S. income level may have different growth rates if they are in different point in the transition to the S.S. (China, Singapore).

In the real world, we see savings rate $\downarrow$ and population growth rate $\downarrow$. What's the implication to the growth rate?

6. Technological Change in the Solow Model

$$Y = F(K, T \times L)$$

$T \times L$ Called effective unites of labor

T represents technological progress. Labor augmenting tech.

- Technology improves at a constant rate, θ .

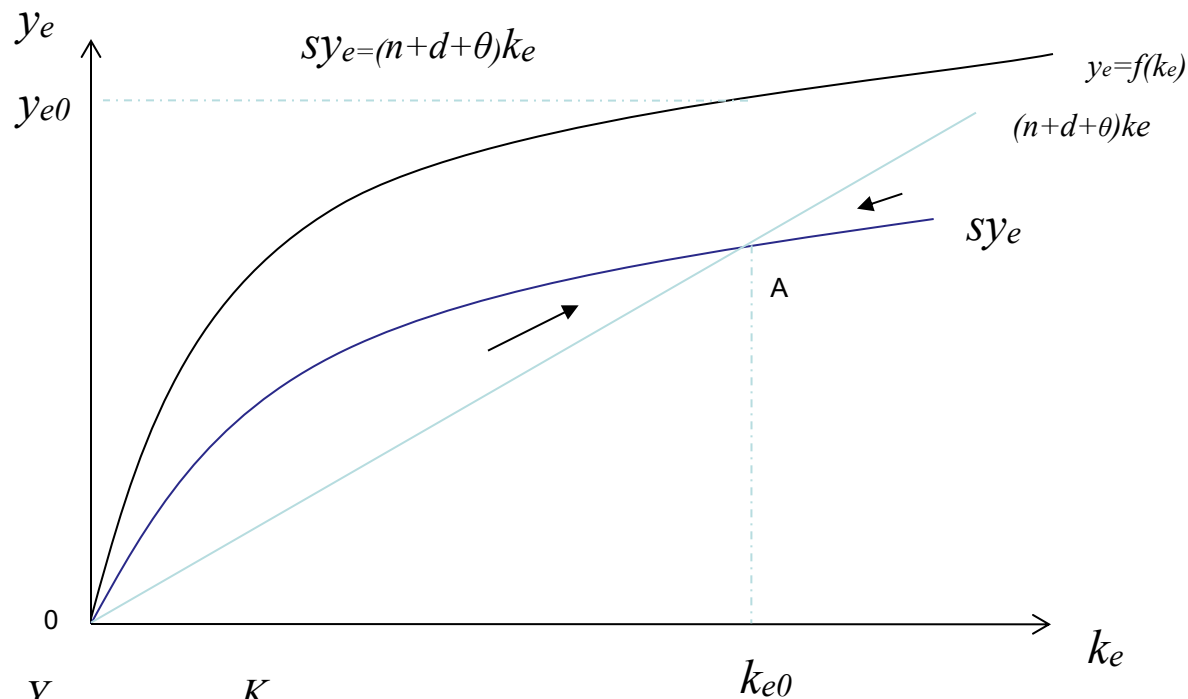
$$\frac{\Delta T}{T} = \theta$$

- The growth of effective labor: $n + \theta$

$$\frac{d(TL)}{TL} = \frac{LdT + TdL}{TL} = dT/T + dL/L = \theta + n$$

n is the rate of labor force growth.

$$\Delta k_e = sy_e - (d + n + \theta)k_e \quad k_e = \frac{K}{TL} \quad y_e = \frac{Y}{TL}$$



- $y_e = \frac{Y}{TL}$ $k_e = \frac{K}{TL}$ constant per effective worker

In the s.s.:

- Y grows at the rate $n + \theta$, effective labor grows at $n + \theta$.
 y_e does not grow,
 y , output per actual worker, grow at θ .
 – Technology changes can cause long-run per capita output growth.

$$\frac{Y}{L} \uparrow \text{ while } \frac{Y}{TL} \text{ fixed}$$

7. Growth rate of output

$$Y = F(K, L, R, A)$$

$R = \text{resources (land, other)}, A = \text{productivity}$

$$\Delta Y = F_K \Delta K + F_L \Delta L + F_R \Delta R + F_A \Delta A, \quad F_K = mp_K, \quad F_L = mp_L,$$

$$\frac{\Delta Y}{Y} = F_K \frac{\Delta K}{Y} + F_L \frac{\Delta L}{Y} + F_R \frac{\Delta R}{Y} + F_A \frac{\Delta A}{Y}$$

$$\Delta K = I \quad \frac{\Delta Y}{Y} = F_K \frac{I}{Y} + F_L \frac{L}{Y} \frac{\Delta L}{L} + F_R \frac{R}{Y} \frac{\Delta R}{R} + F_A \frac{\Delta A}{Y} \quad \left(\frac{\Delta Y}{Y} = F_K \frac{K}{Y} \frac{\Delta K}{K} + F_L \frac{L}{Y} \frac{\Delta L}{L} + F_R \frac{R}{Y} \frac{\Delta R}{R} + F_A \frac{A}{Y} \frac{\Delta A}{A} \right)$$

$$\frac{\Delta Y}{Y} = a + a_K \frac{\Delta K}{K} + a_L \frac{\Delta L}{L} + a_R \frac{\Delta R}{R}$$

Total factor productivity

Where $a \equiv F_A \frac{\Delta A}{Y}$ is productivity growth, “residual” .

$a_K \equiv \frac{F_K K}{Y}$ is Capital share in output.

$a_L \equiv \frac{F_L L}{Y}$ is Labor share in output.

$a_R \equiv \frac{F_R R}{Y}$ is Land share in output.

- Example: $a = 0.02$

$$a_K = 0.2, \quad \frac{\Delta K}{K} = 0.05$$

$$a_L = 0.7, \quad \frac{\Delta L}{L} = 0.03$$

$$a_R = 0.1, \quad \frac{\Delta R}{R} = 0.01$$

- $\frac{\Delta Y}{Y} = ?$

$$\begin{aligned} \frac{\Delta Y}{Y} &= 0.02 + 0.2 \times 0.05 + 0.7 \times 0.03 \times 0.1 \times 0.01 \\ &= 0.02 + 0.01 + 0.021 + 0.001 \\ &= 0.052 = 5.2\% \end{aligned}$$

- This kind of analysis is called growth accounting or sources of growth analysis.

- Shortcomings:
 - The neoclassical growth theory does not provide a satisfactory explanation of changes in technology. “ A ” is usually assumed to be fixed. Since “ A ” determines the growth rate (based on empirical analyses). Economic growth seems to have fallen on countries from heaven.
 - Based on neoclassical growth theory, capital should move from rich country to poor country, because in the poor country labor is abundant and rate of return on capital is high. Evidence does not support this.
 - Also, if more capital moves to the poor countries, the poor countries should grow faster and eventually catch up the rich. In fact, poor countries grew slower.

VI. The New Growth Theory

- Paul Romer (1986), “*Increasing returns & Long-run economic growth*”, Journal of Political Economy.
 - Same labor, more capital→ diminishing returns. But profit – maximizing firms seek out new technologies that economize on labor.
 - Productive knowledge has spill over effect, i.e., new technology of one firm can leak to other firms.
- The spillover of knowledge causes increasing returns to capital.
- Capital moves to the place with more capital. Rich becomes richer.
- Solow takes technology progress as given, or determined by the progress of basic sciences.

- Robert E. Lucas, Jr. (1988), “*The Mechanism of Economic Development*,” Journal of Monetary Economics

- The endogenous growth theories try to explain the technological progress, the “residual”.

$$y = AK^{\alpha} L^{\beta} R^{\gamma} H^{\delta}$$

- H is human capital (through education or learning by doing).
- Countries with more human capital grow faster.
- Human capital has external effects. The more human capital a country has, the higher the productivity of human capital will be.
- Highly skilled workers complement one another.
- Lucas: “what can be paying Manhattan or downtown Chicago rents for, if not for being near other people?”

- Robert E. Lucas, Jr. (1990), “*Why Doesn't Capital Flow from Rich to Poor Countries?*” The American Economic Review
 - Return to physical capital depends positively on the stock of human capital
 - With the consideration of human capital, one can understand why capital does not move to the poor countries.
 - Without considering human capital, India r could be 58 times higher.

- Human capital v.s. knowledge
 - Human capital includes productive skills carried by human being.
 - Knowledge is understanding, familiarity, or information gained by learning or experience. Knowledge can be stored in books, films,
- Barro (1990)
 - Government can affect economic growth. Public infrastructure (roads, schools, hospital, communication systems, health care facilities) can be included in the growth equation to explain the Solow residual.

VII. Some Empirical Findings on Determinants of Economic Growth

Douglass North and Lance Davis (1970), *Institutional Change and American Economic Growth*; North and Thomas (1973), *The Rise of the Western World: A New Economic History*. Cambridge, U.K.: Cambridge U. Press.

- *Institutions* are key to understanding why some countries developed earlier than others.

Daron Acemoglu, Simon Johnson, James Robinson (2001),
“*The Colonial Origins of Comparative Development: An Empirical Investigation*,” *American Economic Review*

- Countries in which economic agents are better protected against the risk of expropriation, and indicator of the quality of institutions, have a higher level of economic development as measured by log GDP per capita.

John Luke Gallup, Jeffrey D. Sachs, and Andrew D. Mellinger,
Geography and Economic Development, *International Regional Science
Review* (1999),

- Two geographic factors influencing growth: distance from the equator and geographical isolation
- Rich countries tend to be located in temperate zones, poor countries in the tropics:
 - tropical disease (malaria, sleeping sickness),
 - intense heat makes hard labor (farming, construction) difficult,
 - rainfall volatility affects agricultural productivity and flooding.
- Landlocked areas tend to be underdeveloped:
 - lower population density means less challenges and cooperation in using resources,
 - lack of access to trade and outside influences (e.g., high transportation cost).

Jeffrey D. Sachs and Andrew M. Warner (1997), Natural resource abundance and economic growth.

- One of the surprising features of modern economic growth is that economies abundant in natural resources have tended to grow slower than economies without substantial natural resources.
- Economies with a high ratio of natural resource exports to GDP in 1970 tended to grow slowly during the subsequent 20-year period 1970-1990.
- This negative relationship holds true even after controlling for many variables found to be important for economic growth by previous authors.
- In the 17th century, resource-poor Netherlands eclipsed Spain. In the 19th and 20th centuries, resource-poor countries such as Switzerland and Japan surged ahead of resource abundant Russia. In the past 30 years, the world's star performers have been the resource-poor Korea, Taiwan, Hong Kong, Singapore -- while the oil-rich countries of Mexico, Nigeria, and Venezuela, have gone bankrupt.

- One early explanation of the phenomenon is social: that easy riches lead to sloth. French political philosopher Jean Bodin [1576, reprinted 1962] claimed:
 - *“men of a fat and fertile soil, are most commonly effeminate and cowards; whereas contrariwise a barren country make men temperate by necessity, and by consequence careful, vigilant, and industrious.”*
- The Authors’ hypothesis:
 - high natural resource abundance leads to increased rent-seeking, corruption, and poorer overall government efficiency;
 - encourages countries to pursue protectionist, state-led development strategies.

- Factors Affecting Economic Development

1. Investment in Physical Capital

2. Investment in Human Capital

3. Openness (Export)

4. Government Size: $\frac{G}{GDP}$

5. Population Growth

6. Fiscal & Monetary Policies

7. Resource Allocation

8. Political Stability

9. Culture

10. Geography

- To grow fast, every part of the economy must function well.

VII. How Long it will Take for a Poor Country Catch Up with a Rich

China and the US:

Total GDP

In 2011, total GDP was \$7.298 trillion in China based on the official exchange rate, and it was \$15.094 trillion in the United States.

t : the number of years it will take to for China to catch the United States

γ : the growth rate of GDP in China

ϕ : the growth rate in the United States

$$7.298e^{t\gamma}=15.094e^{t\phi}$$

US growth rate ϕ	3%	3%
China growth rate γ	8%	7%
Time needed to catch up t	14.5 years	18.17 years
Year	2026	2030

Per capita GDP

In 2011, per capita GDP was \$5,432 trillion in China based on the official exchange rate, and \$48,442 trillion in the United States.

t : the number of years it will take to for China to catch the United States

γ : the growth rate of GDP in China

ϕ : the growth rate in the United States

$$5,432 = 48,442 e^{t\phi}$$

US growth rate ϕ	3%	3%
China growth rate γ	8%	7%
Time needed to catch up t	43.76 years	54.7 years
Year	2055	2066

GDP, Per Capita GDP, Population in UK and China 1500-1900

Year		1500	1600	1700	1820	1850	1870	1890	1900
Total	UK	2,815	6,007	10,709	36,232	63,342	100,180	150,269	184,861
GDP	China	61,800	96,000	82,800	228,600	247,200	189,740	205,304	218,074
	China/UK	21.95	15.98	7.73	6.31	3.9	1.89	1.37	1.18
Per Capita	UK	714	974	1,250	1,706	2,330	3,190	4,009	4,492
GDP	China	600	600	600	600	600	530	540	545
	China/UK	0.84	0.62	0.48	0.35	0.26	0.17	0.13	0.12
Population	UK	3,942	6,170	8,565	21,239	27,181	31,400	37,485	41,155
	China	103,000	160,000	138,000	381,000	412,000	358,000	380,000	400,000
	China/UK	26.13	25.93	16.11	17.94	15.16	11.4	10.14	9.72

In *Wealth Of Nations*, Adam Smith suggested that the wealth of nations be measured in per capita terms. Today when it is said, for example, that England is wealthier than China, it is based the per capita income.

Maddison, 2003, **The World Economy: Historical Statistics**, OECD, Paris 2003
 (1990 International Geary-Khamis Dollars)

Case Study: South Korea and Argentina

- Two countries compared:
 - In 2011, the population was 41 millions in Argentina and 50 millions in South Korea.
 - In 2016, GDP per capita was \$18,494 in Argentina, and \$28,539 in South Korea. In 1960, these numbers were \$1148 vs \$158.
 - S. Korea was a Japanese colony until 1945, wholly dependent on maintaining goodwill of the US for defense against N. Korea.
- Stage of growth
 - South Korea
 - It was not included in Rostow's original book. India was picked by Rostow for takeoff.
 - Investment ratio as income: 15% in 1965, 37% in 1990, 40% in 2000-2007 period.
 - Maturity is attained in less than 60 years, due to difference in productivity gap.
 - Argentina
 - Rostow dated Argentina's precondition for takeoff as an extended period before 1914 and takeoff began in WWI, concluded that "in Latin America the takeoff has been completed in two major cases (Mexico and Argentina)."
 - Argentina had negative growth rate throughout the 1965-1990 period. In the 1980s, domestic investment shrank at a -8.3% rate.
 - Argentina has demonstrated that development progress is not irreversible and that sustained growth can come to an end.

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- Neoclassical theory
 - South Korea
 - The government makes extensive use of development planning, uses a wide range of tax breaks and incentives to induce firms to follow government directives and interventions, sets individual company export targets, orchestrates efforts in various industries to upgrade the average technological level, coordinates foreign technology licensing agreements, uses monopoly power to get the best deal from competing multinationals, and induces firms to move rapidly up the ladder of comparative advantage.
 - Argentina
 - Government policy consistently seemed to support privileged interests rather than broad goals of development. Government failure was usually worse than market failure in the country.
 - Faulty interventionist restrictions, inefficient state enterprise, bias against production for exports, and unnecessary red tape hurt industry and entrepreneurship.